THURSDAY, AUGUST 20 4:00–5:00 P.M.

Lab 2: From a draft file to an analysis dataset

Commission the work, interrogate the evidence, and decide what survives

The policy brief

You are preparing evidence for this question:

> **How is national income associated with under-five mortality across economies between 2000 and 2022?**

The finished dataset must contain one observation per economy–year, GDP per capita (PPP), under-five mortality, a natural-log measure of income, and the economy information needed for interpretation. Observations without usable income or mortality measures do not enter this first analysis.

Work from the RStudio project, this handout, and `code/lab-2-starter.R`. The script is an audit instrument: it exposes evidence and evaluates a candidate output, but it does not build the final pipeline for you.

Two uses of Codex

	Codex as verifier	**Codex as pipeline operator**
Assignment	Examine the available files and make evidence-backed claims.	Choose, implement, and run a complete construction strategy.
Risk	A correct calculation can support an unjustified explanation.	A pipeline can run, save, and pass structural checks while embodying a bad decision.
Your responsibility	Reproduce consequential claims and challenge unsupported inferences.	Inspect the source choice, code, output, and tests before accepting the result.

Codex output is a candidate contribution to the analysis. Fluency, successful execution, and a passing test are evidence of different things; none settles the substantive judgment by itself.

Round I: Codex as verifier

Before inspecting either file, specify the standard by which you will judge a candidate dataset. At minimum, state the unit of observation, identifying variables, period, required measures, exclusions, and one downstream result that a record-level error could change.

Checkpoint 1. State the analytical risk before seeing the records

Describe one plausible data problem and trace its consequences through a briefing product:

$$\text{record-level problem} \longrightarrow \text{changed table, plot, or estimate} \longrightarrow \text{changed policy interpretation.}$$

Run the **evidence audit** in `code/lab-2-starter.R`. Use its output to decide which observations deserve investigation and which comparisons are needed to resolve them.

Then give Codex this verification assignment:

```
Act as a skeptical data reviewer for a policy briefing on national income and
under-five mortality across economies from 2000 to 2022.

Read the handout, data/lab-2/health-briefing-draft.csv,
data/derived/math-camp-wdi-2000-2022.csv, and
data/documentation/indicator-dictionary.csv. Work read-only.

Assess whether the draft is fit for the stated analysis. Examine its unit of
observation, economy-year key, coverage, required variables, missingness,
ranges, and disagreements with the documented project data. For every
conclusion, report the file, variable, economy-year when relevant, and observed
value. Distinguish what the files establish from your explanation of how a
problem may have arisen. End with: (1) your recommendation, (2) the two claims
that matter most for the briefing, and (3) reproducible R checks for them.

Do not edit, deduplicate, or rebuild any file.
```

Checkpoint 2. Adjudicate the verification report

Select the most consequential claim in the response. Reproduce its evidence with R, then record a verdict: *accept*, *revise*, or *reject*.

Claim	
Observed evidence	
What the agent inferred	
Verdict and consequence for the briefing	

Identify one sentence in the response that is more certain than its evidence. If you find none, write the strongest counterargument to the recommendation.

Round II: Codex as pipeline operator

Now commission the complete construction. The assignment deliberately states the analytical requirements without prescribing every implementation choice. The agent must make those choices visible so that you can judge them.

```
Build and run the analysis-data pipeline for this policy question:
How is national income associated with under-five mortality across economies
between 2000 and 2022?

Read the project documentation and relevant data files. Decide which input is
authoritative and justify that decision from project evidence.

Create code/02-health-agent-pipeline.R. It must save:
  outputs/02-health-analysis.csv
  outputs/02-health-build-record.csv

Requirements for the analysis file:
- one row per iso3c + year;
- years 2000 through 2022;
- iso3c, country, region, current income classification, year, GDP per capita
  (PPP), under-five mortality, and log income;
- exclude missing or nonpositive income and missing mortality;
- calculate log income as the natural logarithm of GDP per capita (PPP).

You may create or edit only code/02-health-agent-pipeline.R and the two named
outputs. Do not modify source data, setup files, the starter audit, or the
independent verifier. Run the pipeline. Report the input chosen, exclusions,
files changed, checks performed, and any analytical decision that remains
contestable.
```

Checkpoint 3. Review the commission before accepting execution

What has been delegated? What decision remains yours? Name one way the agent could satisfy the file specification yet produce a substantively indefensible dataset.

Inspect the new script and the agent's report. Do not evaluate the final output only from the prose summary. Look for the chosen input, any rule that resolves multiple records, sample restrictions, transformation, and actual files written.

Audit the pipeline, not merely its exit status

Run the **candidate-output audit** in `code/lab-2-starter.R`. It reports two distinct forms of evidence:

1. **structural evidence**: required columns, unique keys, period, missingness, ranges, and recomputation of logged income; and
2. **source concordance**: missing or unexpected keys and values that disagree with the documented project data.

Checkpoint 4. Find the strongest reason to accept or reject the pipeline

Does the script implement the stated unit and sample?

Does the output pass structural checks?

Does the output agree with documented source records?

Is every conflict-resolution rule defensible without relying on row order?

Decision: accept, revise, or reject

A pipeline can pass one set of checks and fail another. If everything passes, identify a consequential question that the current audit still cannot answer.

From checked data to a briefing table

Use the candidate only after you have accepted it or revised the pipeline. A policy colleague now asks:

> **In 2022, how did typical income and under-five mortality differ across current World Bank income groups?**

Write an R pipeline that starts from `outputs/02-health-analysis.csv`, restricts the data to 2022, groups by `income_level_current`, and creates one row per income group with:

- the number of economies;
- median GDP per capita (PPP); and
- median under-five mortality.

Arrange the table from lowest to highest median mortality and save it as `outputs/02-health-briefing-by-income.csv`. This is an unweighted descriptive comparison: each economy contributes once to its group's 2022 summary.

Checkpoint 5. Defend the grouped briefing result

Before computing, state how the unit changes from the analysis file to the briefing table and how a repeated economy–year could alter a group median or count. After computing, confirm that the group counts sum to the number of distinct economies observed in 2022.

Write two briefing sentences: one numerical comparison supported by the table and one limitation that prevents a causal or historical-classification claim.

Ask Codex to verify the grouped result without rewriting it: independently recompute the counts and medians from the accepted analysis file, compare them with the saved briefing table, and identify any sentence that goes beyond the evidence.

Checkpoint 6. Distinguish the two agent roles

In no more than four sentences, explain:

1. what Codex contributed as a verifier;
2. what it contributed as a pipeline operator;
3. one mistake or overclaim you caught—or the most credible failure you attempted to falsify; and
4. whether you would release the dataset and briefing table, and why.

Keep the prompt, generated pipeline, audit evidence, and your decision. The reusable skill is not following a prescribed sequence. It is commissioning analytical work precisely and knowing what evidence would make you reject it.